

TE Curve Verification Pipeline Multi-Model Curve Comparison Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report compares representative `TE Curve Verification Pipeline` `TE` curves by overlaying multiple model predictions on the same original measured curve. The plots are intended to show whether each model tracks the local harmonic oscillations rather than only the broad mean trend.

Scope

- each comparison image contains four deterministic held-out test curves;
- forward comparisons are shown on forward curves only;
- backward comparisons are shown on backward curves only;
- Wave 1 screening keeps the three strongest family-best models by `Curve MAE [deg]` within each direction;
- `Original Curve` uses the same visual weight as predictions and a dark-gray color for balanced comparison.

Metrics Summary

Forward Wave52r Integrated Specialist Trained Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52r_integrated_a02_seed_314159	wave52r_integrated_specialist_trained	global	0.001400	0.001658	2.856
wave52r_integrated_a03_seed_161803	wave52r_integrated_specialist_trained	global	0.001403	0.001662	2.862
wave52r_integrated_a04_seed_161803	wave52r_integrated_specialist_trained	global	0.001398	0.001655	2.848
wave52r_integrated_a05_seed_161803	wave52r_integrated_specialist_trained	global	0.001400	0.001658	2.852
wave52r_integrated_a06_seed_271828	wave52r_integrated_specialist_trained	global	0.001404	0.001662	2.858
wave52r_integrated_a07_seed_314159	wave52r_integrated_specialist_trained	global	0.001403	0.001661	2.858
wave52r_integrated_a07_seed_161803	wave52r_integrated_specialist_trained	global	0.001405	0.001664	2.862

Backward Wave52r Integrated Specialist Trained Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52r_integrated_a02_seed_314159	wave52r_integrated_specialist_trained	global	0.001523	0.001811	2.923
wave52r_integrated_a03_seed_161803	wave52r_integrated_specialist_trained	global	0.001512	0.001801	2.909
wave52r_integrated_a04_seed_161803	wave52r_integrated_specialist_trained	global	0.001517	0.001802	2.907
wave52r_integrated_a05_seed_161803	wave52r_integrated_specialist_trained	global	0.001521	0.001808	2.916
wave52r_integrated_a06_seed_271828	wave52r_integrated_specialist_trained	global	0.001523	0.001809	2.919
wave52r_integrated_a07_seed_314159	wave52r_integrated_specialist_trained	global	0.001523	0.001810	2.919
wave52r_integrated_a07_seed_161803	wave52r_integrated_specialist_trained	global	0.001521	0.001809	2.918

Forward Wave52r Offline Leader Cross Surface Promotion Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52r_promotion_h08_fw_seed_161803	wave52r_offline_leader_cross_surface_promotion	Fw	0.001688	0.001985	3.477

Backward Wave52r Offline Leader Cross Surface Promotion Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52r_promotion_k01_global_seed_271828	wave52r_offline_leader_cross_surface_promotion	global	0.001523	0.001811	2.923

Forward Accepted Non PINN Incumbent Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
accepted_periodic_gru_sequence_Fw	accepted_non_pinn_incumbent	Fw	0.001618	0.001931	3.278
accepted_periodic_mlp_harmonic_Fw	accepted_non_pinn_incumbent	Fw	0.001694	0.002008	3.439

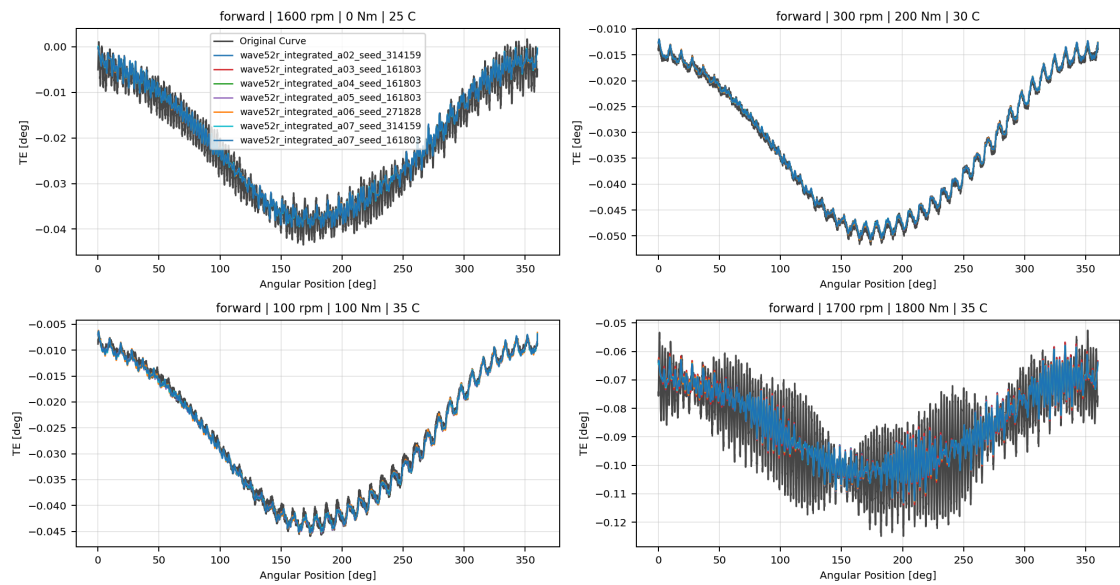
Backward Accepted Non PINN Incumbent Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
accepted_periodic_gru_sequence_Bw	accepted_non_pinn_incumbent	Bw	0.001837	0.002196	3.494
accepted_periodic_mlp_harmonic_Bw	accepted_non_pinn_incumbent	Bw	0.001912	0.002263	3.581

Comparison Gallery - Forward Wave52r Integrated Specialist Trained Overlay

Included models: wave52r_integrated_a02_seed_314159 , wave52r_integrated_a03_seed_161803 , wave52r_integrated_a04_seed_161803 , wave52r_integrated_a05_seed_161803 , wave52r_integrated_a06_seed_271828 , wave52r_integrated_a07_seed_314159 , wave52r_integrated_a07_seed_161803 .

Forward Wave52r Integrated Specialist Trained Overlay

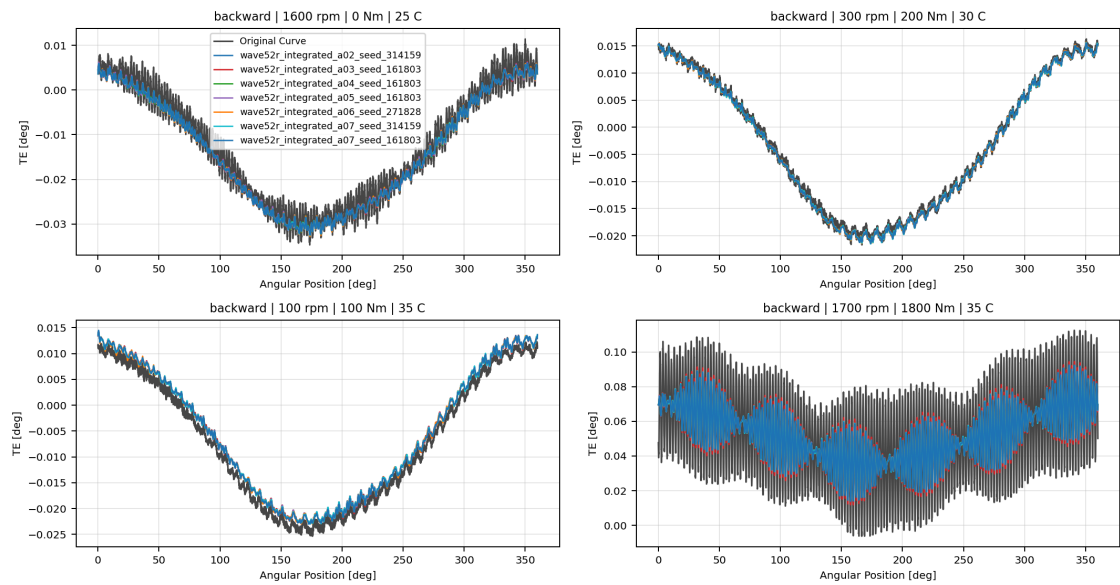


Forward Wave52r Integrated Specialist Trained Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave52r Integrated Specialist Trained Overlay

Included models: wave52r_integrated_a02_seed_314159 , wave52r_integrated_a03_seed_161803 , wave52r_integrated_a04_seed_161803 , wave52r_integrated_a05_seed_161803 , wave52r_integrated_a06_seed_271828 , wave52r_integrated_a07_seed_314159 , wave52r_integrated_a07_seed_161803 .

Backward Wave52r Integrated Specialist Trained Overlay

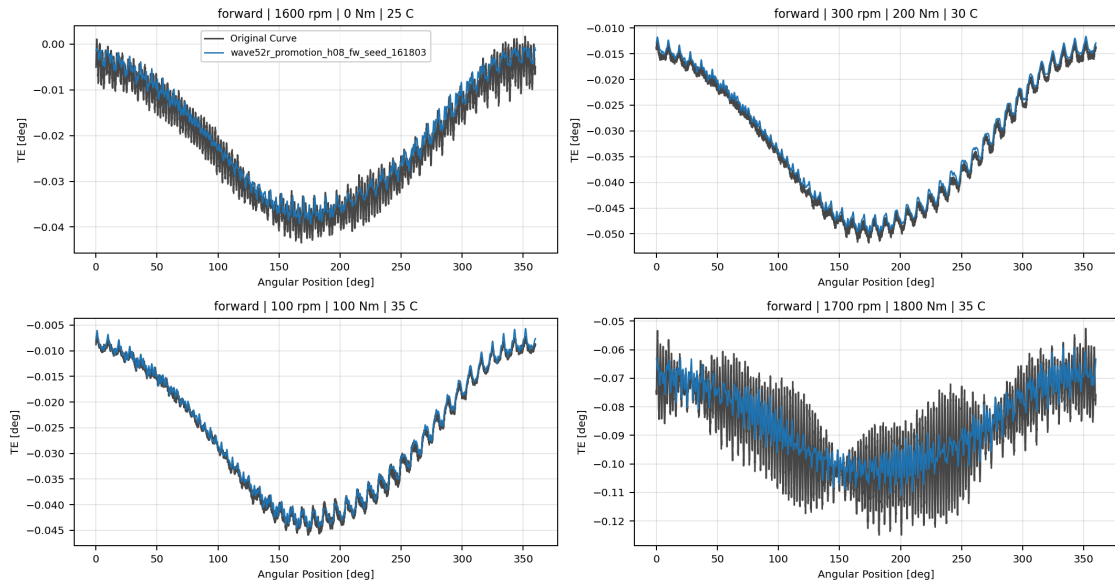


Backward Wave52r Integrated Specialist Trained Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave52r Offline Leader Cross Surface Promotion Overlay

Included models: `wave52r_promotion_h08_fw_seed_161803` .

Forward Wave52r Offline Leader Cross Surface Promotion Overlay

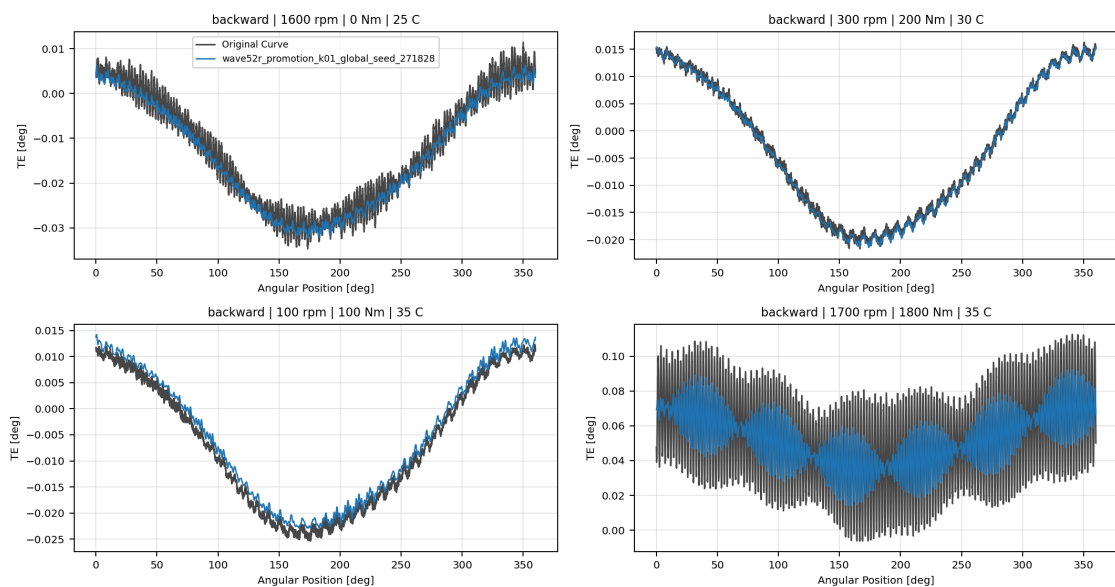


Forward Wave52r Offline Leader Cross Surface Promotion Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave52r Offline Leader Cross Surface Promotion Overlay

Included models: `wave52r_promotion_k01_global_seed_271828` .

Backward Wave52r Offline Leader Cross Surface Promotion Overlay

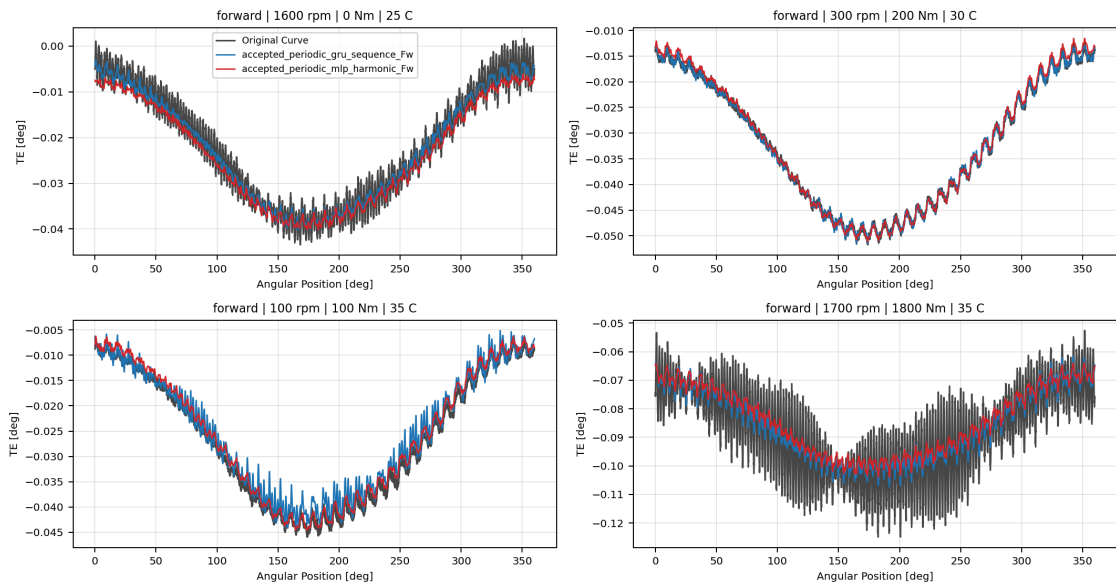


Backward Wave52r Offline Leader Cross Surface Promotion Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Accepted Non PINN Incumbent Overlay

Included models: `accepted_periodic_gru_sequence_Fw` , `accepted_periodic_mlp_harmonic_Fw` .

Forward Accepted Non PINN Incumbent Overlay

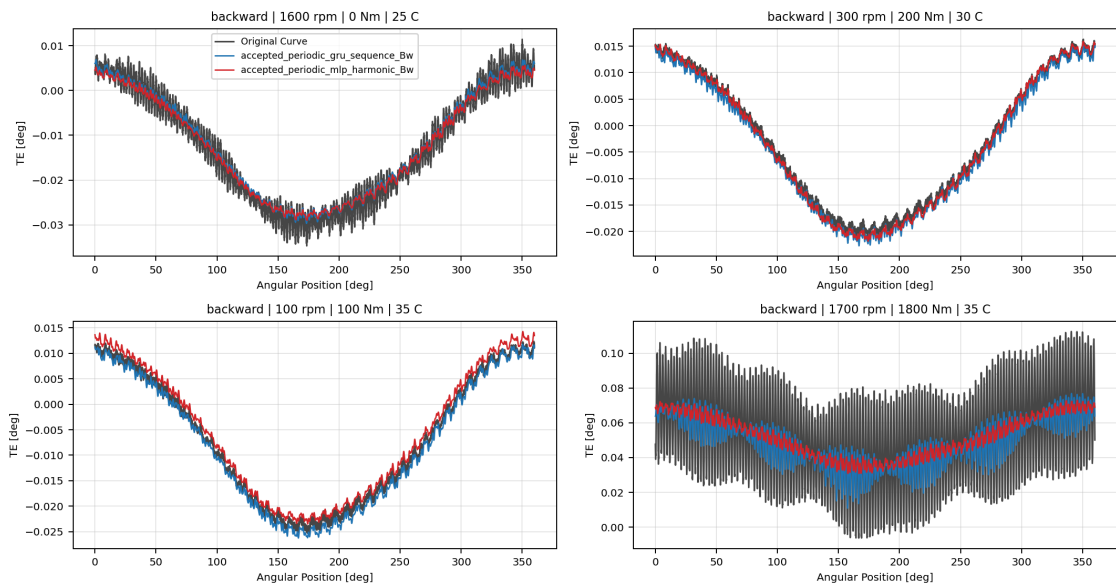


Forward Accepted Non PINN Incumbent Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Accepted Non PINN Incumbent Overlay

Included models: `accepted_periodic_gru_sequence_Bw` , `accepted_periodic_mlp_harmonic_Bw` .

Backward Accepted Non PINN Incumbent Overlay



Backward Accepted Non PINN Incumbent Overlay TE Curve Verification Pipeline comparison

Output Artifacts

- output directory:
`output\validation_checks\wave52r_integrated_specialist_track2_multi_model_curve_comparison_report\2026-08-04-00-39-20__track2_multi_model_curve_comparison_report ;`
- summary YAML:
`output\validation_checks\wave52r_integrated_specialist_track2_multi_model_curve_comparison_report\2026-08-04-00-39-20__track2_multi_model_curve_comparison_report\track2_multi_model_curve_comparison_summary.yaml ;`
- metrics CSV:
`output\validation_checks\wave52r_integrated_specialist_track2_multi_model_curve_comparison_report\2026-08-04-00-39-20__track2_multi_model_curve_comparison_report\track2_multi_model_curve_comparison_metrics.csv ;`
- report Markdown:
`doc\reports\analysis\te_curve_verification_pipeline\02_visual_reports\wave52r_integrated_specialist_multi_model_curve_comparison_report\[2026-08-04]\track2_multi_model_curve_comparison_report.md .`